

Part A: Classifying Numbers

Classify each number using all that apply: rational, irrational, whole, integer.

1. $\sqrt{18}$

13. $-\frac{8}{11}$

2. -19

14. $\sqrt{49}$

3. $-\frac{1}{9}$

15. 18

4. $-6\frac{1}{2}$

16. $-5\frac{1}{3}$

5. $\sqrt{16}$

17. $-1\frac{7}{9}$

6. $\sqrt{100}$

18. $\sqrt{17}$

7. $1\frac{5}{11}$

19. $\sqrt{11}$

8. 3

20. $\sqrt{13}$

9. $-3\frac{1}{4}$

21. $-4\frac{1}{2}$

10. 5

22. $\sqrt{12}$

11. $\sqrt{6}$

23. $\sqrt{3}$

12. $\sqrt{15}$

24. $\sqrt{4}$

25. 15

37. 16

26. -2

38. $-2\frac{1}{4}$

27. $-\frac{3}{11}$

39. $\frac{2}{5}$

28. $\sqrt{14}$

40. $\sqrt{5}$

29. -3

41. -1

30. -10

42. -6

31. -20

43. $\sqrt{2}$

32. $-\frac{1}{3}$

44. $\sqrt{121}$

33. $-1\frac{4}{5}$

45. $-1\frac{8}{11}$

34. $\frac{8}{9}$

46. $\sqrt{8}$

35. $2\frac{2}{5}$

47. 13

36. 20

48. 11

49. $\frac{5}{9}$

50. $\sqrt{144}$

58. $-6\frac{2}{3}$

51. $-1\frac{6}{7}$

59. -13

52. 19

60. -11

53. $8\frac{1}{2}$

61. -9

54. -5

62. -18

55. $\sqrt{19}$

63. $-1\frac{5}{6}$

56. $4\frac{2}{3}$

64. -12

57. -8

65. $\sqrt{9}$

66. $-\frac{1}{11}$

Part B: Negative Number Operations

Evaluate each expression completely.

1. $10 + (-8)$

5. $9 + 9$

2. $8 + 8$

6. $-12 - (-4)$

3. $-5 + (-1)$

7. $-3 - (-1)$

4. $-8 - 10$

8. $-10 - 11$

9. $-7 - 4$

22. $-15 + 11 \cdot 12$

10. $2 \cdot (-4)$

23. $14 + (-9) \cdot (-3)$

11. $2 \cdot (-2)$

24. $5 + 8 \cdot (-5)$

12. $-8 \cdot 8$

25. $8 + 2 \cdot 9$

13. $2 \cdot (-7)$

26. $36 \div (-4) - (-7)$

14. $35 \div (-7)$

27. $(2 + (-17))^2 \div (-3)$

15. $-21 \div 3$

28. $(-9)^2 - 13 \cdot 6$

16. $7 \cdot (-5)$

29. $-15 - 3 \cdot ((-6) + (-15))$

17. $-7 \cdot 2$

30. $5 - 3 \cdot (4 + 5)$

18. $8 \cdot (-2)$

31. $8^3 - (-2) \cdot 3$

19. $-19 + 7 \cdot 2$

32. $(4 + (-10))^2 \div 1$

20. $15 + (-5) \cdot 3$

33. $(-9 + (-4))^2 \div (-1)$

21. $-14 + (-4) \cdot (-2)$

34. $-7 - 12 \cdot ((-2) + (-7))$

Part C: Decimals

Evaluate. Write each answer as a decimal.

1. $3.3 + 18.3$

6. 3.9×2.5

2. $19.1 - 13.3$

7. $99.178 \div 1.7$

3. $10.1 - 0.2$

8. $70.38 \div 1.7$

4. 6.5×4.1

9. $84.96 \div 1.6$

5. 9.5×4.8

Part D: Decimals to Fractions*Write each decimal as a fraction in lowest terms.*

1. 4.5

2. 3.45

Part E: Fractions to Decimals*Write each fraction as a decimal.*

1. $\frac{37}{4}$

2. $\frac{69}{10}$

Part F: Decimals to Percents*Write each decimal as a percent.*

1. 0.424

2. 0.444

Part G: Percents to Decimals*Write each percent as a decimal.*

1. 54.3%

2. 74.8%

Part H: Fractions*Evaluate. Write every answer in lowest terms.*

1. $\frac{1}{2} - \frac{1}{5}$

7. $\frac{4}{5} \cdot \frac{1}{2}$

2. $\frac{1}{5} - \frac{2}{3}$

8. $\frac{1}{2} \div \frac{3}{5}$

3. $\frac{5}{6} - \frac{1}{4}$

9. $\frac{2}{3} \cdot \frac{1}{2}$

4. $\frac{2}{3} + \frac{1}{2}$

10. $4\frac{2}{3} - 2\frac{1}{5}$

5. $\frac{1}{3} + \frac{2}{3}$

11. $3 - 3$

6. $\frac{4}{5} \cdot \frac{1}{3}$

12. $2\frac{5}{9} + 2\frac{2}{9}$

13. $2\frac{5}{6} + 5\frac{2}{3}$

Part I: Writing Fractions in Lowest Terms

Write each fraction in lowest terms.

1. $\frac{20}{16}$

3. $\frac{28}{12}$

2. $\frac{49}{56}$

4. $\frac{66}{24}$

Part J: Exponent Rules

Simplify. Write answers with positive exponents.

1. $x^5 \cdot 3x^3$

2. $2x^7 \cdot x^2$

3. $4x^2 \cdot x^7$

15. $\frac{2x^{10}}{x^7}$

4. $3x^7 \cdot 2x^6$

16. $\frac{3x^8}{x^6}$

5. $4x^5 \cdot 3x^7$

17. $\frac{3x^9}{x^6}$

6. $3x^5 \cdot 3x^5$

18. $\frac{x^8}{x^4}$

7. $4x^2 \cdot x^5$

19. $\frac{4x^4}{x^2}$

8. $x^5 \cdot 4x^2$

20. $\frac{4x^6}{x^2}$

9. $4x^2 \cdot x^4$

10. $x^4 \cdot x^2$

21. $\frac{x^{10}}{x^5}$

11. $x^3 \cdot 4x^7$

22. $\frac{x^{13}}{x^7}$

12. $\frac{4x^9}{x^3}$

23. $(6x^6)^4$

13. $\frac{2x^4}{x^3}$

24. $(3x^8)^2$

14. $\frac{2x^9}{x^2}$

25. $(4x^7)^3$

26. $(9x^3)^4$

27. $(5x^2)^3$

40. $8 \cdot (2^1)^{-1}$

28. $(2x^6)^2$

41. $4 \cdot 10^{-4}$

29. $(6x^6)^2$

42. $\frac{9}{4^{-4}}$

30. $(6x^3)^3$

43. $\frac{7}{7^{-4}}$

31. $(5x^3)^2$

44. $6 \cdot (2^3)^{-4}$

32. $(4x^6)^2$

45. $6^5 \cdot 6^{-2}$

33. $(5x^5)^2$

46. $2^4 \cdot 2^0$

34. $3 \cdot 3^{-2}$

47. $5^2 \cdot 5^{-2}$

35. $3 \cdot (5^2)^{-3}$

48. $7^3 \cdot 7^{-2}$

36. $\frac{4}{4^{-2}}$

49. $2^5 \cdot 2^{-1}$

37. $\frac{2}{3^{-1}}$

50. $5^3 \cdot 5^{-3}$

38. $2 \cdot 6^{-1}$

51. $5^4 \cdot 5^{-2}$

39. $4 \cdot 6^{-1}$

52. $3^4 \cdot 3^{-1}$

53. $6^5 \cdot 6^{-3}$

60. $(4x^{-3})^3 \cdot x^{11}$

54. $7^5 \cdot 7^{-1}$

61. $(3x^{10}y^9)^2 \cdot 11x^{12}y^5$

55. $2^2 \cdot 2^{-4}$

62. $\frac{10x^{12}y^6}{2x^5y^3}$

56. $\left(\frac{3x^7}{3x^5}\right)^2$

63. $(12x^2y^7)^3 \cdot 3x^{12}y^1$

57. $\left(\frac{2x^9}{x^2}\right)^2$

64. $\left(\frac{5x^{11}}{x^5}\right)^2$

58. $\frac{3x^{10}y^8}{x^8y^3}$

65. $(10x^{12}y^4)^3 \cdot 3x^8y^4$

59. $(5x^{-1})^2 \cdot x^{11}$

66. $(4x^{-2})^3 \cdot x^{11}$

Answer Key

Part A: Classifying Numbers

- | | | |
|------------------------------|------------------------------|------------------------------|
| 1. Irrational | 23. Irrational | 45. Rational |
| 2. Rational, Integer | 24. Rational, Integer, Whole | 46. Irrational |
| 3. Rational | 25. Rational, Integer, Whole | 47. Rational, Integer, Whole |
| 4. Rational | 26. Rational, Integer | 48. Rational, Integer, Whole |
| 5. Rational, Integer, Whole | 27. Rational | 49. Rational |
| 6. Rational, Integer, Whole | 28. Irrational | 50. Rational, Integer, Whole |
| 7. Rational | 29. Rational, Integer | 51. Rational |
| 8. Rational, Integer, Whole | 30. Rational, Integer | 52. Rational, Integer, Whole |
| 9. Rational | 31. Rational, Integer | 53. Rational |
| 10. Rational, Integer, Whole | 32. Rational | 54. Rational, Integer |
| 11. Irrational | 33. Rational | 55. Irrational |
| 12. Irrational | 34. Rational | 56. Rational |
| 13. Rational | 35. Rational | 57. Rational, Integer |
| 14. Rational, Integer, Whole | 36. Rational, Integer, Whole | 58. Rational |
| 15. Rational, Integer, Whole | 37. Rational, Integer, Whole | 59. Rational, Integer |
| 16. Rational | 38. Rational | 60. Rational, Integer |
| 17. Rational | 39. Rational | 61. Rational, Integer |
| 18. Irrational | 40. Irrational | 62. Rational, Integer |
| 19. Irrational | 41. Rational, Integer | 63. Rational |
| 20. Irrational | 42. Rational, Integer | 64. Rational, Integer |
| 21. Rational | 43. Irrational | 65. Rational, Integer, Whole |
| 22. Irrational | 44. Rational, Integer, Whole | 66. Rational |

Part B: Negative Number Operations

- | | | |
|-----------|-----------|------------|
| 1. 2 | 13. -14 | 25. 26 |
| 2. 16 | 14. -5 | 26. -2 |
| 3. -6 | 15. -7 | 27. -75 |
| 4. -18 | 16. -35 | 28. 3 |
| 5. 18 | 17. -14 | 29. 48 |
| 6. -8 | 18. -16 | 30. -22 |
| 7. -2 | 19. -5 | 31. 518 |
| 8. -21 | 20. 0 | 32. 36 |
| 9. -11 | 21. -6 | 33. -169 |
| 10. -8 | 22. 117 | 34. 101 |
| 11. -4 | 23. 41 | |
| 12. -64 | 24. -35 | |

Part C: Decimals

- | | | |
|---------|----------|----------|
| 1. 21.6 | 4. 26.65 | 7. 58.34 |
| 2. 5.8 | 5. 45.6 | 8. 41.4 |
| 3. 9.9 | 6. 9.75 | 9. 53.1 |

Part D: Decimals to Fractions

1. $\frac{9}{2}$

2. $\frac{69}{20}$

Part E: Fractions to Decimals

1. 9.25

2. 6.9

Part F: Decimals to Percents

1. 42.4%

2. 44.4%

Part G: Percents to Decimals

1. 0.543

2. 0.748

Part H: Fractions

1. $\frac{3}{10}$

2. $-\frac{7}{15}$

3. $\frac{7}{12}$

4. $1\frac{1}{6}$

5. 1

6. $\frac{4}{15}$

7. $\frac{2}{5}$

8. $\frac{5}{6}$

9. $\frac{1}{3}$

10. $2\frac{7}{15}$

11. 0

12. $4\frac{7}{9}$

13. $8\frac{1}{2}$

Part I: Writing Fractions in Lowest Terms

1. $1\frac{1}{4}$

2. $\frac{7}{8}$

3. $2\frac{1}{3}$

4. $2\frac{3}{4}$

Part J: Exponent Rules

1. $3x^8$

2. $2x^9$

3. $4x^9$

4. $6x^{13}$

5. $12x^{12}$

6. $9x^{10}$

7. $4x^7$

8. $4x^7$

9. $4x^6$

10. x^6

11. $4x^{10}$

12. $4x^6$

13. $2x$

14. $2x^7$

15. $2x^3$

16. $3x^2$

17. $3x^3$

18. x^4

19. $4x^2$

20. $4x^4$

21. x^5

22. x^6

23. $1296x^{24}$

24. $9x^{16}$

25. $64x^{21}$

26. $6561x^{12}$

27. $125x^6$

28. $4x^{12}$

29. $36x^{12}$

30. $216x^9$

31. $25x^6$

32. $16x^{12}$

33. $25x^{10}$

34. $\frac{1}{3}$

35. $\frac{3}{15625}$

36. 64

37. 6

38. $\frac{1}{3}$

39. $\frac{2}{3}$

40. 4

41. $\frac{1}{2500}$

42. 2304

43. 16807

44. $\frac{3}{2048}$

45. 216

46. 16

47. 1

48. 7

49. 16

50. 1

51. 25

52. 27

53. 36

54. 2401

55. $\frac{1}{4}$

56. x^4

57. $4x^{14}$

58. $3x^2y^5$

59. $25x^9$

60. $64x^2$

61. $99x^{32}y^{23}$

62. $5x^7y^3$

63. $5184x^{18}y^{22}$

64. $25x^{12}$

65. $3000x^{44}y^{16}$

66. $64x^5$